

Araav Rotella

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EDUCATION

University of Maryland

B.S. in Computer Science, Minor in Mathematics

College Park, MD

Expected May 2027

- **GPA:** 3.5/4.0 | Dean's List (3 Semesters) | Coursework: Algorithms, Programming Languages, Data Science, Machine Learning, OOP, Linear Algebra

EXPERIENCE

AI/ML Intern

Allegis Group

May 2025 – Present

Hanover, MD

- **Clario:** Built and deployed a **5,100+ line full-stack platform** (Streamlit + Snowflake Cortex) across 10 features, shipping via GitHub Actions CI/CD to Azure App Service with Checkmarx SAST/SCA security gates.
- Implemented a **RAG chatbot** orchestrating 14 Cortex AI endpoints over 7 Cortex Search object types to generate personalized career guidance and skill-gap analysis.
- Built **role-based access** with Microsoft SSO enabling a manager coaching view, plus AI headshot generation via Google Vertex AI and session summaries across 5+ Snowflake tables.
- **Ingestion Pipeline:** Ingested **716+ job files** with zero data loss via a Snowflake stored procedure using CORTEX.PARSE.DOCUMENT with page-splitting and row-per-page flattening.
- Achieved parallel ingestion across **14 concurrent sessions** with a queue-driven orchestration layer (RUNNING/COMPLETE/FAILED tracking), eliminating 4 production-blocking bugs.
- **ML/NLP:** Cut annual costs **\$3.5K** migrating the Win Probability Model to a Snowflake ML pipeline; raised F1 from 70% to 80% via Bayesian hyperparameter optimization.
- Eliminated **83% of false positives** in message filtering with a content-moderation model, reaching 100% true-positive detection across 12M+ rows using Snowflake Cortex AI.

PROJECTS

Pocket Vogue (Full-Stack AI Stylist Microservice)

June 2025

- Architected an end-to-end **FastAPI** microservice on Supabase (PostgreSQL + Object Storage), persisting CLIP-extracted labels and CDN image URLs across a multi-endpoint REST API.
- Engineered a multimodal pipeline using **CLIP (ViT-B/32)** to run 4 concurrent zero-shot classification tasks per image, automating item tagging at upload.
- Built a **SHA-256** deduplication layer that fingerprints uploads to intercept redundant images, cutting model-inference calls and storage writes.
- Shipped an O(1) set-difference analytics endpoint surfacing wardrobe coverage gaps; deployed the backend on **Railway** with a React swipe-card frontend on Vercel.

AI Resume Editor (Snowflake Cortex + LaTeX)

March 2025 - May 2025

- Built a full-stack editor parsing PDFs with **PARSE_DOCUMENT**, tailoring content to a JD via Cortex AI.COMPLETE, and rendering a LaTeX template to downloadable, ATS-parsable PDF.
- Implemented **5 AI editing modes** (refine, tailor, custom edit, changes report, skill extraction) with side-by-side diffs and PDF/text export.

LLM From Scratch

September 2025

- Built a **transformer-based LLM** from scratch in Python/NumPy/PyTorch, implementing tokenization, positional encodings, multi-head self-attention, and feed-forward layers.
- Trained a 4-layer ~10M-parameter model to **perplexity 120 → 35**; tuned AdamW, LR warmup, and gradient clipping to improve convergence 25%, and explored LoRA and RLHF.

TECHNICAL SKILLS

Languages: Python, Java, SQL, C, Rust, JavaScript, HTML/CSS, MATLAB

ML & AI: PyTorch, TensorFlow, Gymnasium (OpenAI), Scikit-learn, XGBoost, HuggingFace, Pandas, NumPy

Tools & Concepts: Git, Linux, Snowflake, FastAPI, Supabase, React, CI/CD, REST APIs, Reinforcement Learning, LLMs